

Subject Description Form

Subject Code	ITC3219I
Subject Title	Textiles for Intimate Apparel and Activewear
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: SFT322CD Materials & Accessories for Intimate Apparel and Activewear
Objectives	The subject provides the academic and technical knowledge of textile materials with specific reference to their uses in intimate apparel and active wear. It develops understanding of the functional requirements of these products, as well as evaluates and incorporates the latest technical know-how for the design and development of intimate apparel and active wear.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) analyse the characteristics and functional performance of various textile materials for intimate apparel and active wear, integrate the fashion material, latest technologies and technical innovations in product design and product development; (b) critically evaluate the methods of producing various textile materials and accessories in response to the future needs of the international and local intimate apparel and active wear markets; (c) relate the quality characteristics of the various textile materials and accessories to the performance of intimate apparel and active wear products; (d) demonstrate their theoretical and practical knowledge and understanding of intimate apparel or active wear product design and development in undertaking problem-centred assignments, critically analysing the results and information, and professionally communicating and presenting the findings and conclusions.
Subject Synopsis/ Indicative Syllabus	(I) Materials Requirements for Intimate Apparel and Activewear Introduction Ergonomic and utility requisites Tactile sensation Thermal balance and thermal regulation (II) Textile Fibres Used in Intimate and Activewear Natural, regenerated and synthetic fibres Thermoplastic fibres, characteristics and moulding properties Elastomeric fibres, mechanical and chemical properties New developments in fibres and textile materials for intimate and activewear

	(III) Warp Knitting Technology Terminologies of warp knits The structure of warp knits Machine classification - Tricot and Raschel machines Patterning mechanism and basic designing principles Yarn preparation and manufacturing processes Warp knit fabric analysis Double need bar warp knit fabrics (IV) Active Knit Fabrics Functional needs of activewear application The influence of knitted fabrics' structure on the thermal and moisture management properties Double face knitted fabrics (V) Lingerie Laces Types, design and properties of lace for intimate apparel Introduction of multi-guide bar Raschel machine Karl Mayer machines for lingerie laces Development of lace machines Lace design process and effects of different lace patterns on the aesthetic and performance of intimate apparel Manufacturing processes for both warp knit and woven lace types (VI) Embroidery Types, designs and properties of embroidered fabrics used in intimate apparel design Type of backing used for embroidered fabrics Design details used in embroidery Type of embroidery machines Manufacturing processes used in embroidery (VII) Narrow Fabric / Elastic Trimming Types, constructions and properties of narrow elastic fabrics Production process – weaving/ knitting, dyeing and finishing Functional innovations in narrow fabric elastic tapes Aesthetic innovations in narrow fabric elastic tapes (VIII) Innovative Accessories for Intimate Apparel and Activewear Underwires Hook-and-eye tapes Fasteners Bones and supportive materials Thermo-adhesive tape use in activewear
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Teaching/Learning Methodology	Fundamental concepts and knowledge will be introduced through lectures. Guest seminars given by professionals will enable students to gain insights into industry practice. Tutorial classes will be used to supplement formal lectures and involve more interactive teaching and learning. Laboratory sessions will supplement the theory and concepts supplied in the lectures and facilitate the understanding of the structure of warp knit fabrics and laces. Factory and exhibition visits will also be arranged. The assessment will include problem-centred assignments, factory visit reports, laboratory reports and/or problem solving activities. For example, after a factory visit, students will be required to prepare a written report and/or complete problem-centred assignment to reflect their understanding, critical thinking and problem-solving techniques. Guidelines and/or well-designed unstructured questions may be given in advance so that students will focus on certain areas and demonstrate their understanding of subject matter.																																																			
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>1. Fabric analysis</td><td>15%</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Factory visit reports</td><td>20%</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>3. Tests</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Fabric analysis assessment will be used to assess students’ ability to analyse the characteristics and functional performance of various textile materials. Factory visit reports will be used to assess whether the students can critically evaluate the processes involved in the manufacture of various textile materials and accessories in response to the future needs of the international and local intimate apparel and active wear markets. Tests will assess all fundamental concepts and knowledge taught in this subject.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	Continuous Assessment	50%	✓	✓	✓	✓	1. Fabric analysis	15%	✓				2. Factory visit reports	20%		✓		✓	3. Tests	15%	✓	✓	✓	✓	Examination	50%	✓	✓	✓	✓	Total	100%				
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3. Tests	15%	✓	✓	✓	✓																																															
Examination	50%	✓	✓	✓	✓																																															
Total	100%																																																			

	The exam will assess all fundamental concepts and knowledge taught in this subject, particularly the characteristics, functional performance and production of various textile materials and accessories used for intimate apparel and active wear.	
Student Study Effort Expected	Class contact:	
	• Lecture	26 Hrs.
	• Tutorial	12 Hrs.
	Other student study effort:	
	• Assignments	60 Hrs.
	• Factory visit reports	8 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	<u>Books</u> Au, K. F. (2011), <i>Advances in Knitting Technology</i>. Woodhead Publishing Series in Textiles. Hu, J. and Lu, J. (2015), “Chapter 3 Recent developments in elastic fibers and yarns for sportswear”. <i>Textiles for Sportswear</i>, Shishoo, R. (eds), Woodhead Publishing Series in Textiles. Yip, J. (2016) “Chapter 1 Advanced textiles for intimate apparel”, <i>Advances in women’s intimate apparel technology</i>, Yu, W. M. (eds), Woodhead Publishing in association with the Textile Institute. Yip, J. (2016) “Chapter 2 Narrow fabric elastic tapes”, <i>Advances in women’s intimate apparel technology</i>, Yu, W. M. (eds), Woodhead Publishing in association with the Textile Institute. Yip, J. (2016) “Chapter 3 Innovative accessories for intimate apparel”, <i>Advances in women’s intimate apparel technology</i>, Yu, W. M. (eds), Woodhead Publishing in association with the Textile Institute.	

Supplementary

Kanakaraj, P., Ramachandran, R. (2015), “Active knit fabrics – functional needs of sportswear application”, *Journal of Textile and Apparel, Technology and Management*, Vol 9, Iss 2, p.1-11.

Yip, J., Ng, S. P. (2009), “Study of Three-dimensional Spacer Fabrics: Moulding Properties for Intimate Apparel Application”, *Journal of Materials Processing Technology*, 209, pp.58-62.

Yip, J., Ng, S. P. (2008), “Study of Three-dimensional Spacer Fabrics: Physical and Mechanical Properties”, *Journal of Materials Processing Technology*, 206, pp.359-364.

Periodicals

Sia A. (2014+), Intimate apparel journal New York
<http://library.polyu.edu.hk/search/t?SEARCH=intimate+apparel+journal&searchscope=2>

Journal of Sport and Health Science
<http://library.polyu.edu.hk/search~S2/?searchtype=t&searcharg=Journal+of+Sport+and+Health+Science&searchscope=2&sortdropdown=-&SORT=D&extended=0&SUBMIT=Search&searchlimits=&searchorigarg=tintimate+apparel+journal>